

mathematics of probability called stochastic processes. The notion of path, as opposed to outcome, indicates that it is not a mere MBA-style scenario analysis, but the examination of a sequence of scenarios along the course of time. We are not just concerned at where a bird can end up tomorrow night, but rather at all the various places it can possibly visit during the time interval. We are not concerned with what the investor's worth would be in, say, a year, but rather of the heart-wrenching rides he may experience during that period. The word *sample* stresses that one sees only one realization among a collection of possible ones. Now a sample path can be both deterministic or random, which brings the next distinction.

A *random sample path*, also called a random run, is the mathematical name for such succession of virtual historical events, starting at a given date and ending at another, except that they are subjected to some varying level of uncertainty. However, the word random should not be mistaken for equiprobable (i.e. having the same probability). Some outcomes will give a higher probability than others. An example of a random sample path can be the body temperature of your explorer cousin during his latest bout with typhoid fever, measured hourly from the beginning to the end of his episode. It can also be a simulation of the price of your favorite technology stock, measured daily at the close of the market, over, say, one year. Starting at \$100, in one scenario it can end up at \$20 having seen a high of \$220, in another it can end up at \$145 having seen a low of \$10. Another example is the evolution of your wealth during an evening at a casino. You start with \$1000 in your pocket, and measure it every 15 minutes. In one sample path you have \$2200 at midnight; in another you barely have \$20 left for a cab fare.

Stochastic processes refer to the dynamics of events unfolding with the course of time. Stochastic is a fancy Greek name for random. This branch of probability concerns itself with the study of the evolution of successive random events – one could call it the mathematics of history. The key about a process is that it has time in it.

What is a Monte Carlo generator? Imagine that you can replicate a perfect roulette wheel in your attic without having recourse to a carpenter. Computer programs can be written to simulate just about anything. They are even better (and cheaper) than the roulette wheel built by your carpenter, as this may be inclined to favor one number

more than others owing to a possible slant in its build or the floor of your attic. These are called the biases.

Monte Carlo simulations are closer to a toy than anything I have seen in my adult life. One can generate thousands, perhaps millions of random sample paths, and look at the prevalent characteristics of some of their features. The assistance of the computer is instrumental in such studies. The glamorous reference to Monte Carlo indicates the metaphor of simulating the random events in the manner of a virtual casino. One sets conditions believed to resemble the ones that prevail in reality, and launches a collection of simulations around possible events. With no mathematical literacy we can launch a Monte Carlo simulation of an 18-year-old Christian Lebanese playing successively Russian roulette for a given sum, and see how many of these attempts result in enrichment, or how long it takes on average before he hits the obituary. We can change the barrel to contain 500 holes, a matter that would decrease the probability of death, and see the results.

Monte Carlo simulation methods were pioneered in martial physics in the Los Alamos laboratory during the A bomb preparation. They became popular in financial mathematics in the 1980s, particularly in the theories of the random walk of asset prices. Clearly, we have to say that the example of Russian roulette does not need such apparatus, but many problems, particularly those resembling real-life situations, require the potency of a Monte Carlo simulator.

MONTE CARLO MATHEMATICS

It is a fact that “true” mathematicians do not like Monte Carlo methods. They believe that they rob us of the finesse and elegance of mathematics. They call it “brute force”. For we can replace a large portion of mathematical knowledge with a Monte Carlo simulator (and other computational tricks). For instance, someone with no formal knowledge of geometry can compute the mysterious, almost mystical Pi. How? By drawing a circle inside of a square, and “shooting” random bullets into the picture (as in an arcade), specifying equal probabilities of hitting any point on the map (something called a uniform distribution). The ratio of bullets inside the circle divided by those inside and outside the circle will